Axolabs GmbH

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TABLE OF CONTENTS

1 RELATED DOCUMENTS **4**

2 DEFINITIONS AND ABBREVIATIONS **5**

3 ACTIVITIES TO BE PERFORMED **6**

3.1 Summary **8**

3.2 Limited Assay Qualification **9**

4 RESULTS FROM QUALIFICATION RUNS IN CELL CULTURE MEDIA **9**

5 ASSAY PROCEDURE **27**

5.1 Preparation of Stock Solution **27**

5.1.1 Instruments and Materials 27

5.1.2 Reagents 28

5.1.3 Method 28

5.2 Preparation of the Assay Protocol **28**

5.3 Preparation of Working Solutions **28**

5.3.1 Instruments and Materials 28

5.3.2 Reagents 29

5.3.3 Method 29

5.4 Preparation of Cell Culture Media Lysate and Hybridization **29**

5.4.1 Instruments and Materials 29

5.4.2 Reagents 29

5.4.3 Method 29

5.4.3.1 Sample Dilution 31

5.5 HPLC Analysis **31**

5.5.1 Material and Reagents 31

5.5.2 Instruments 32

5.5.3 Method 32

5.6 Evaluation of HPLC Chromatograms **33**

6 RECORDS **34**

7 CHANGE HISTORY **34**

TABLE OF TABLES

Table 1 Parameters of final method 8

Table 2 Calibration Curve Accuracy in cell culture media 13

Table 3 Confirmation of LLOQ in cell culture media 15

Table 4 Carry-over in cell culture media 16

Table 5 Accuracy and Precision in cell culture media 17

Table 6 Short term stability cell culture media 19

Table 7 Short term stability in cell culture media (addition of 0.1% SDS to working solutions) 22

Table 8 Short term stability in cell culture media: Addition of 0.1% SDS to workings solution and to stability samples 24

Table 9 Dilution Integrity in cell culture media 26

Table 10 Gradient table 33

TABLE OF FIGURES

Figure 1 Structure and sequence of the analyte 2439-exNA 6

Figure 2 HPLC Profile of 2439-exNA in cell media culture. Chromatograms at the LLOQ (1 ng/mL, upper picture) and the ULOQ (1,000 ng/mL, lower picture) are shown. 7

1 RELATED DOCUMENTS

This instruction describes the execution of a PNA-HPLC-Assay based on the following documents:

- KU-RGL-SOP-002 (formerly: REG-102) – Methoden-Validierung im GLP/GCP-Bereich (*Method Validation in GLP/GCP Area*) with the following subordinate Instructions, Forms, Spreadsheet Applications:
  - KU-RGL-ESS-002 (formerly: REG-102-S02) – Method Validation Data Report
  - KU-RGL-ESS-005 (formerly: REG-102-S05) – Method Validation Protocol PNA Assay - Liquid Matrices - OnePot
- KU-RGL-SOP-009 (formerly: REG-109) – Durchführung des PNA-HPLC-Assay (*Execution of the PNA-HPLC-Assay*) with the following subordinate Instructions, Forms, Spreadsheet Applications:
  - KU-RGL-ESS-013 (formely: REG-109-S03) – PNA Assay Instruction and Protocol – Liquid Matrices - OnePot
  - KU-RGL-ESS-014 (formerly: REG-109-S04) – Preparation of HPLC Buffers & Stock Solutions
  - KU-RGL-INS-005 (formerly: REG-109-I01) – PNA-HPLC-Assay SST
- KU-RGL-SOP-010 (formerly: REG-110) – Herstellung von Standardlösungen mit einer Präzisionswaage (*Preparation of Standard Solutions Using a Precision Balance*) with the following subordinate Instructions, Forms, Spreadsheet Applications:
- Development protocol in the project folder AN400.0-24 under the following link:

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The above mentioned documents describe the assay execution under nonGLP-relevant conditions. The parameters and results described in this document are generated from a nonGLP method development, which was conducted following the above mentioned SOPs.

2 DEFINITIONS AND ABBREVIATIONS

°C	Grad Celsius
AB0X	Analytical batch AN400.0-24_04_AB0X
ACN	Acetonitrile
AEX-HPLC	Anion-Exchange High-Performance Liquid Chromatography
CV	Coefficient of Variation
DNA	Deoxyribonucleic Acid
ed	Edition
EDTA	Ethylenediaminetetraacetic Acid
FDA	Food and Drug Administration
GLP	Good Laboratory Practice
HPLC	High Performance Liquid Chromatography
ICH	International Conference on Harmonization
IFU	Instructions For Use
LC/MS	Liquid Chromatography / Mass Spectrometry
LLOQ	Lower Limit of Quantitation
min	Minute
n.a.; N/A; N.A.	Not applicable
PBS	Phosphate Buffered Saline
PI	Principal Investigator
PNA	Peptide Nucleic Acid
QC	Quality Control
rcf	Relative centrifugal force
RNA	Ribonucleic Acid
RT	Room temperature
SD	Standard Deviation
SDS	Sodium Dodecyl Sulfate
siRNA	Small Interfering Ribonucleid Acid
SOP	Standard Operating Procedure
SST	System Suitability Test(ing)
t	Timepoint
ULOQ	Upper Limit of Quantitation

WS	Working Solution

3 ACTIVITIES TO BE PERFORMED

This method development report describes an anion exchange (AEX)-HPLC method with fluorescence detection that allows the specific detection of the analyte 2439-exNA from cell culture media. 2439-exNA is a dimer composed of two identical siRNAs, which are linked at their sense strands with a tetraethylene glycol linker, with the sequence depicted in Figure 1:

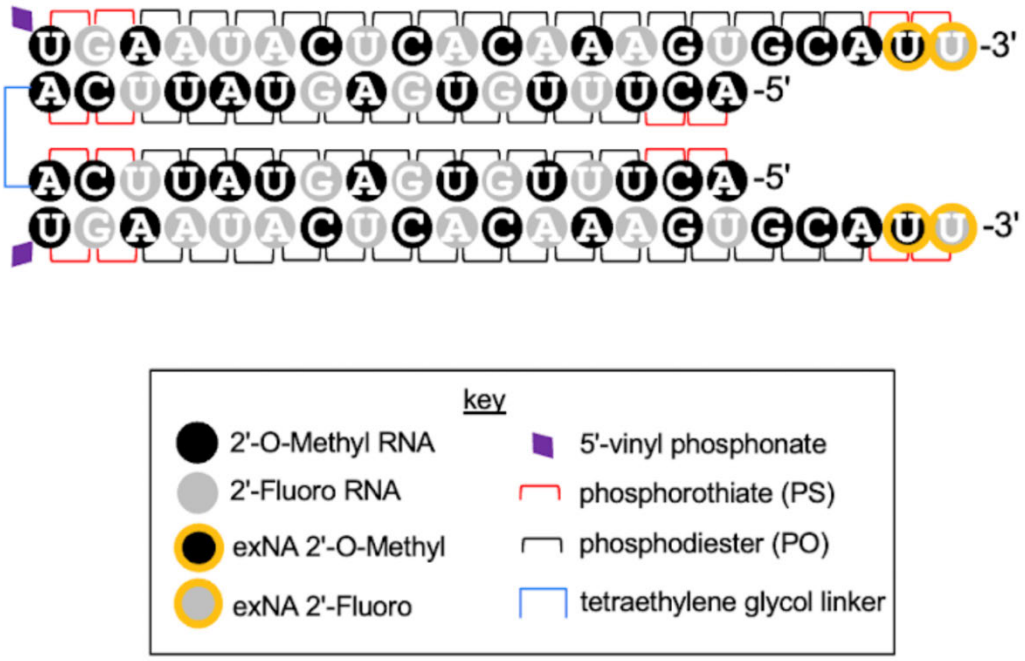

Figure 1 Structure and sequence of the analyte 2439-exNA

The assay is based on the specific hybridization of the antisense strands of 2439-exNA to the complementary PNA-probe. The PNA-probe is N-terminally conjugated with an Atto425-fluorescent dye. The utilized PNA-BroadInst-01 is complementary to the antisense strand of 2439-exNA and has the following sequence:

N-term Atto425-OO-cdc ttt gtg dgt d C-term

This method was developed without having any metabolites of 2439-exNA

Figure 2 shows the HPLC-profile of 2439-exNA in cell culture media.

[redacted]

3.1 Summary

[redacted]

[redacted]	[redacted]	[redacted]
[redacted]	[redacted]	[redacted]
[redacted]	[redacted]	[redacted]
[redacted]	[redacted]	[redacted]
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[redacted]	[redacted]	[redacted]
[redacted]	[redacted]	[redacted]

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[redacted]

[redacted]

4 Results from Qualification runs in cell culture media

In this section, the raw data for the determination of calibration curve accuracy (Table 2), confirmation of LLOQ (Table 3), carry-over (Table 4), accuracy and precision (Table 5), short term stability (Table 6 Table 7, Table 8) and dilution integrity (Table 9) from 2439-exNA in cell culture media are shown. The results were generated in two qualifying runs. Calibration standard and QC samples were prepared by spiking 2439-exNA into cell culture media. Samples were prepared according to the assay procedure described in Section 5.

During the qualification runs, the calibration range was between 1.0 ng/mL and 1,000.0 ng/mL. All calibration standards were within ±15 % (±20 % at the LLOQ) of the nominal concentration.

Sensitivity

The LLOQ was confirmed at 1.0 ng/mL in cell culture media. No signal was detected in corresponding blank cell culture media samples and the 2439-exNA peaks at the LLOQ of 1.0 ng/mL were at least 5 times the background signal in the analysis run. At the LLOQ, the mean back-calculated concentration was 100.0 % of nominal with a CV of 8.2 %.

Carry Over

Carry over cell culture media samples were run after the 1,000.0 ng/mL calibration standards. There was no distinct analyte signal detectable in the carry-over samples after the 1,000.0 ng/mL standards.

Accuracy and Precision

QC samples were prepared at 4 concentration levels which are covering the calibration curve range:

- at the LLOQ (QC-LLOQ): 1.0 ng/mL
- at 3 times above the LLOQ (QC-Low): 3.0 ng/mL
- at around 50 % of the calibration curve range (QC-Mid): 450.0 ng/mL
- at least at 75 % of the upper calibration curve range (QC-High): 900.0 ng/mL

The QC samples for the determination of the accuracy and precision of the method met the acceptance criteria with following results:

- 100 % of the QC samples at all concentration levels were within ±15 % (±20 % at the LLOQ) of their nominal value.

AB01

(only QC samples at the 3 concentration levels QC-Low, Mid and High were included in this batch):

- The mean accuracies at the 3 concentration levels were 103.3 % (QC-Low), 98.2 % (QC-Mid) and 101.0 % (QC-High).
- The mean precisions at the 3 concentration levels were 4.6 % (QC-Low), 7.2 % (QC-Mid) and 5.6 % (QC-High).

AB02:

- The mean accuracies at the 4 concentration levels were 106.7 % (QC-LLOQ), 103.3 % (QC-Low), 106.1 % (QC-Mid) and 106.2 % (QC-High).
- The mean precisions at the 4 concentration levels were 5.4 % (QC-LLOQ), 6.8 % (QC-Low), 1.6 % (QC-Mid) and 1.0 % (QC-High).

AB03:

(only QC samples at the 3 concentration levels QC-Low, Mid and High were included in this batch):

- The mean accuracies at the 3 concentration levels were 96.7 % (QC-Low),102.9 % (QC-Mid) and 101.8 % (QC-High).
- The mean precisions at the 3 concentration levels were 0.0 % (QC-Low),1.7 % (QC-Mid) and 1.6 % (QC-High).

Short-Term-Stability

The short term stability of 2439-exNA in cell culture media was successfully tested for up to 4 hours. Aliquots of the 2439-exNA at the 2 QC concentration levels QC-Low and QC-High were stored on wet ice and at room temperature for 4 hours and analyzed in triplicate

During the preparation of the stability samples, the working solution was prepared by diluting the stock solution with cell culture medium in glass vials and in LoBind tubes. The working solution was then spiked into the blank cell medium in a 1:20 ratio (working solution: medium). Preparation of samples was followed by an incubation step for t=0 (control) and t=4h. Afterwards, samples were further processed just like study samples and analyzed.

With this approach, the control samples at t=0 and t=4h did not meet acceptance criteria of ±15% of the nominal value regarding accuracy. Since the concentration after 4 h incubation did not change significantly, it indicates that the analyte is stable but during sample preparation, the desired amount of analyte could not be spiked into the sample. The most likely reason for this is an unspecific binding effect of the analyte to the surface of the vials and pipette tips.

In order to prevent this binding effect, working solutions were prepared using cell culture medium containing 0.1% SDS. The working solution was then spiked into the sample in a 1:20 ratio (working solution:medium), resulting in a final SDS concentration of stability samples of 0.005% SDS.

The stability samples prepared using working solution with SDS met the acceptance criteria regarding accuracy, both at t=0 and at =4 h.

See Table 6 and Table 7 for all results for the stability sample test.

After 4 hours at room temperature the mean back-calculated concentrations were at 85.6 % (QC-Low) and 102.4% (QC-High) of the nominal value.

After 4 hours on wet ice the mean back-calculated concentrations were at 96.7 % (QC-Low) and 101.8% (QC-High) of the nominal value.

Furthermore the stability samples were tested without the addition of any other reagent, and were analysed after addition of 1 volume percent of 10% SDS (SDS was added after preparation). Results see Table 8. Due to binding effects during dilution series, the results did not meet the acceptance criteria. However, the peak areas of the untreated and SDS-treated sample differ (results see Table 6 and Table 8). The peak areas for the SDS-treated samples were higher compared to untreated, indicating that binding effects can be reversed/reduced by addition of SDS.

Dilution Integrity

The precision and accuracy of cell culture media sample dilution was evaluated using a DI-QC standard prepared at concentration levels 1,000-fold (500,000 ng/mL) above the ULOQ of 1,000.0 ng/mL. This QC sample was analyzed and quantified after 1:1,000 dilution in 3 replicates. Samples were diluted with different buffer solution in advance (see below and section 5.4.3.1).

Dilution with 4 M Urea:

- The mean accuracy was 97.7 % and the CV was 9.6 %

Dilution with cell medium buffer

- The mean accuracy was 87.7 % and the CV was 10.3 %

Dilution with cell medium buffer/ 1 % Tween/ 1 mM TCEP

- The mean accuracy was 91.8 % and the CV was 4.8 %

The Dilution Integrity test with LC-MS water did not meet the acceptance criteria.

Table 2 Calibration Curve Accuracy in cell culture media MCM+

2439-exNA		Cal-Level-2		Cal-Level-3		Cal-Level-4		Cal-Level-5		Cal-Level-6		Cal-Level-7	
		1.0 ng/mL		5.0 ng/mL		50.0 ng/mL		100.0 ng/mL		500.0 ng/mL		1,000.0 ng/mL	
Run ID	Sample No.	Calc. 2439-exNA Conc. ng/mL	Acc. criterion +/-20 % of Nominal	Calc. 2439-exNA Conc. ng/mL	Acc. criterion +/-15 % of Nominal	Calc. 2439-exNA Conc. ng/mL	Acc. criterion +/-15 % of Nominal	Calc. 2439-exNA Conc. ng/mL	Acc. criterion +/-15 % of Nominal	Calc. 2439-exNA Conc. ng/mL	Acc. criterion +/-15 % of Nominal	Calc. 2439-exNA Conc. ng/mL	Acc. criterion +/-15 % of Nominal
AN400.0-24_04_AB01	4 - 9	0.9	90.0%	4.7	94.0%	50.6	101.2%	103.4	103.4%	505.0	101.0%	1,009.2	100.9%
AN400.0-24_04_AB01	84 - 89	1.1	110.0%	5.1	102.0%	51.4	102.8%	102.1	102.1%	486.9	97.4%	991.5	99.2%
Mean		1.00	100.0%	4.90	98.0%	51.00	102.0%	102.75	102.8%	495.95	99.2%	1,000.35	100.0%
SD (+/-)		0.14		0.28		0.57		0.92		12.80		12.52	
CV (%)		14.1%		5.8%		1.1%		0.9%		2.6%		1.3%	
N		2		2		2		2		2		2	
AN400.0-24_04_AB02	4 - 9	1.0	100.0%	5.0	100.0%	49.3	98.6%	99.1	99.1%	501.4	100.3%	998.9	99.9%
AN400.0-24_04_AB02	33 - 38	1.0	100.0%	5.0	100.0%	50.0	100.0%	98.3	98.3%	499.6	99.9%	1,003.4	100.3%
Mean		1.00	100.0%	5.00	100.0%	49.65	99.3%	98.70	98.7%	500.50	100.1%	1,001.15	100.1%
SD (+/-)		0.00		0.00		0.49		0.57		1.27		3.18	
CV (%)		0.0%		0.0%		1.0%		0.6%		0.3%		0.3%	
N		2		2		2		2		2		2	

Table 2 Calibration Curve Accuracy in cell culture media MCM+_continued

2439-exNA		Cal-Level-2		Cal-Level-3		Cal-Level-4		Cal-Level-5		Cal-Level-6		Cal-Level-7	
		1.0 ng/mL		5.0 ng/mL		50.0 ng/mL		100.0 ng/mL		500.0 ng/mL		1,000.0 ng/mL	
Run ID	Sample No.	Calc. 2439-ex NA Conc. ng/mL	Acc. criterion +/-20 % of Nominal	Calc. 2439-ex NA Conc. ng/mL	Acc. criterion +/-15 % of Nominal	Calc. 2439-ex NA Conc. ng/mL	Acc. criterion +/-15 % of Nominal	Calc. 2439-ex NA Conc. ng/mL	Acc. criterion +/-15 % of Nominal	Calc. 2439-ex NA Conc. ng/mL	Acc. criterion +/-15 % of Nominal	Calc. 2439-ex NA Conc. ng/mL	Acc. criterion +/-15 % of Nominal
AN400.0-24_04_AB03	5 - 10	1.1	110.0%	4.8	96.0%	50.8	101.6%	101.1	101.1%	500.7	100.1%	998.8	99.9%
AN400.0-24_04_AB03	50 - 55	1.0	100.0%	4.8	96.0%	49.4	98.8%	99.4	99.4%	494.3	98.9%	1,005.9	100.6%
Mean		1.05	105.0%	4.80	96.0%	50.10	100.2%	100.25	100.3%	497.50	99.5%	1,002.35	100.2%
SD (+/-)		0.07		0.00		0.99		1.20		4.53		5.02	
CV (%)		6.7%		0.0%		2.0%		1.2%		0.9%		0.5%	
N		2		2		2		2		2		2	

Table 3 Confirmation of LLOQ in cell culture media MCM+

2439-exNA		Cal-Level-1			Cal-Level-2		
		0.0 ng/mL			1.0 ng/mL		
Run ID	Sample No.	Peak Area [mV*min]	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ≤ 20 % of LLOQ	Peak Area [mV*min]	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion +/-20 % of LLOQ
AN400.0-24_04_AB01	3, 4	0.0000	BQL	0.0%	0.0377	0.9	90.0%
AN400.0-24_04_AB01	83, 84	0.0000	BQL	0.0%	0.0453	1.1	110.0%
AN400.0-24_04_AB02	3. 4	0.0000	BQL	0.0%	0.0318	1.0	100.0%
AN400.0-24_04_AB02	32. 33	0.0000	BQL	0.0%	0.0343	1.0	100.0%
AN400.0-24_04_AB03	4, 5	0.0000	BQL	0.0%	0.0443	1.1	110.0%
AN400.0-24_04_AB03	49, 50	0.0000	BQL	0.0%	0.0399	1.0	100.0%
Mean		0.00000	n.a.	0.0%	0.03888	1.02	101.7%
SD (+/-)		0.00000	n.a.		0.00537	0.08	
CV (%)		n.a.	n.a.		13.8%	7.4%	
N		6	6		6	6	

Table 4 Carry-over in cell culture media MCM+

2439-exNA		Cal-Level-2			Cal-Level-7			Carry Over		
		1.0 ng/mL			1,000.0 ng/mL			0.0 ng/mL		
Run ID	Sample No.	Peak Area [mv*min]	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion +/-20 % of Nominal	Peak Area [mv*min]	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion +/-15 % of Nominal	Peak Area [mv*min]	Calculated 2439-exNA Conc. ng/mL	Acc. Criterion ≤ 20 % of LLOQ
AN400.0-24_04_AB01	4, 9, 10	0.0377	0.9	90.0%	45.0502	1,009.2	100.9%	0.0000	BQL	0.0%
AN400.0-24_04_AB01	84, 89, 90	0.0453	1.1	110.0%	44.2594	991.5	99.2%	0.0000	BQL	0.0%
AN400.0-24_04_AB02	4, 9, 10	0.0318	1.0	100.0%	41.7895	998.9	99.9%	0.0000	BQL	0.0%
AN400.0-24_04_AB02	33, 38, 39	0.0343	1.0	100.0%	41.9792	1,003.4	100.3%	0.0000	BQL	0.0%
AN400.0-24_04_AB03	5, 10, 11	0.0443	1.1	110.0%	43.4996	998.8	99.9%	0.0000	BQL	0.0%
AN400.0-24_04_AB03	50, 55, 56	0.0399	1.0	100.0%	43.8048	1,005.9	100.6%	0.0000	BQL	0.0%

Table 5 Accuracy and Precision in cell culture media MCM+

2439-exNA		QC-LLOQ		QC-Low		QC-Mid		QC-High	
		1.0 ng/mL		3.0 ng/mL		450.0 ng/mL		900.0 ng/mL	
Run ID	Sample No.	Calc. 2439-exNA Conc. ng/mL	Acc. Criterion +/-20 % of Nominal	Calc. 2439-exNA Conc. ng/mL	Acc. Criterion +/-15 % of Nominal	Calc. 2439-exNA Conc. ng/mL	Acc. Criterion +/-15 % of Nominal	Calc. 2439-exNA Conc. ng/mL	Acc. Criterion +/-15 % of Nominal
AN400.0-24_04_AB01	12, 13, 14	n.a.	n.a.	3.2	106.7%	419.5	93.2%	873.2	97.0%
	79, 80, 81	n.a.	n.a.	3.0	100.0%	464.2	103.2%	945.5	105.1%
Mean		n.a.	n.a.	3.10	103.3%	441.85	98.2%	909.35	101.0%
SD (+/-)		n.a.		0.14		31.61		51.12	
CV (%)		n.a.		4.6%		7.2%		5.6%	
N		n.a.		2		2		2	
AN400.0-24_04_AB02	15, 18, 21, 24	1.1	110.0%	3.1	103.3%	473.8	105.3%	950.6	105.6%
	16, 19, 22, 25	1.0	100.0%	3.0	100.0%	470.8	104.6%	948.1	105.3%
	17, 20, 23, 36	1.1	110.0%	2.8	93.3%	472.1	104.9%	952.2	105.8%
	42, 43, 44	n.a.	n.a.	3.3	110.0%	484.2	107.6%	956.1	106.2%
	59, 60, 61	n.a.	n.a.	3.3	110.0%	486.9	108.2%	971.2	107.9%
Mean		1.07	106.7%	3.10	103.3%	477.56	106.1%	955.64	106.2%
SD (+/-)		0.06		0.21		7.43		9.17	
CV (%)		5.4%		6.8%		1.6%		1.0%	
N		3		5		5		5	

Table 5 Accuracy and Precision in cell culture media MCM+_continued

2439-exNA		QC-LLOQ		QC-Low		QC-Mid		QC-High	
		1.0 ng/mL		3.0 ng/mL		450.0 ng/mL		900.0 ng/mL	
Run ID	Sample No.	Calc. 2439-exNA Conc. ng/mL	Acc. Criterion +/-20 % of Nominal	Calc. 2439-exNA Conc. ng/mL	Acc. Criterion +/-15 % of Nominal	Calc. 2439-exNA Conc. ng/mL	Acc. Criterion +/-15 % of Nominal	Calc. 2439-exNA Conc. ng/mL	Acc. Criterion +/-15 % of Nominal
AN400.0-24_04_AB03	13, 14, 15	n.a.	n.a.	2.9	96.7%	457.6	101.7%	905.5	100.6%
	45, 46, 47	n.a.	n.a.	2.9	96.7%	468.9	104.2%	926.0	102.9%
Mean		n.a.	n.a.	2.90	96.7%	463.25	102.9%	915.75	101.8%
SD (+/-)		n.a.		0.00		7.99		14.50	
CV (%)		n.a.		0.0%		1.7%		1.6%	
N		0		2		2		2	

Table 6 Short term stability cell culture media MCM+

2439-exNA				QC-Low 3.0 ng/mL		QC-High 900.0 ng/mL	
Run ID	**Date**	**Sample No.**	**Time Point**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion +/-15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion +/-15 % of Nominal**
AN400.0-24_04_AB01_Qualibatch WS in glass vials	27-Jun-2024	16, 19	0 hours	2.0	66.7%	678.1	75.3%
	27-Jun-2024	17, 20		2.1	70.0%	676.6	75.2%
	27-Jun-2024	18, 21		2.0	66.7%	645.4	71.7%
Mean				2.03	67.8%	666.70	74.1%
SD (+/-)				0.06		18.46	
CV (%)				2.8%		2.8%	
N				3		3	
AN400.0-24_04_AB01_Qualibatch WS in glass vials	27-Jun-2024	58, 61	4 hours on ice	2.1	70.0%	656.3	72.9%
	27-Jun-2024	59, 62		2.3	76.7%	603.3	67.0%
	27-Jun-2024	60, 63		2.2	73.3%	628.6	69.8%
Mean				2.20	73.3%	629.40	69.9%
SD (+/-)				0.10		26.51	
CV (%)				4.5%		4.2%	
N				3		3	

Table 6 Short term stability cell culture media MCM+_continued

2439-exNA				QC-Low 3.0 ng/mL		QC-High 900.0 ng/mL	
Run ID	**Date**	**Sample No.**	**Time Point**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion +/-15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion +/-15 % of Nominal**
AN400.0-24_04_AB01_Qualibatch WS in glass vials	27-Jun-2024	37, 40	4 hours at RT	2.0	66.7%	499.9	55.5%
	27-Jun-2024	38, 41		2.3	76.7%	625.4	69.5%
	27-Jun-2024	39, 42		2.2	73.3%	622.8	69.2%
Mean				2.17	72.2%	582.70	64.7%
SD (+/-)				0.15		71.72	
CV (%)				7.1%		12.3%	
N				3		3	
AN400.0-24_04_AB01_Qualibatch WS in LoBind	27-Jun-2024	23, 26	0 hours	2.1	70.0%	772.0	85.8%
	27-Jun-2024	24, 27		2.1	70.0%	221.9	24.7%
	27-Jun-2024	25, 28		2.1	70.0%	316.2	35.1%
Mean				2.10	70.0%	436.70	48.5%
SD (+/-)				0.00		294.18	
CV (%)				0.0%		67.4%	
N				3		3	

Table 6 Short term stability cell culture media MCM+_continued

2439-exNA				QC-Low 3.0 ng/mL		QC-High 900.0 ng/mL	
Run ID	**Date**	**Sample No.**	**Time Point**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion +/-15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion +/-15 % of Nominal**
AN400.0-24_04_AB01_Qualibatch WS in LoBind	27-Jun-2024	65, 68	4 hours on ice	2.1	70.0%	730.8	81.2%
	27-Jun-2024	66, 69		2.1	70.0%	679.4	75.5%
	27-Jun-2024	67, 70		2.2	73.3%	697.3	77.5%
Mean				2.13	71.1%	702.50	78.1%
SD (+/-)				0.06		26.09	
CV (%)				2.7%		3.7%	
N				3		3	
AN400.0-24_04_AB01_Qualibatch WS in LoBind	27-Jun-2024	44, 47	4 hours at RT	2.0	66.7%	694.0	77.1%
	27-Jun-2024	45, 48		2.0	66.7%	697.1	77.5%
	27-Jun-2024	46, 49		1.9	63.3%	605.4	67.3%
Mean				1.97	65.6%	665.50	73.9%
SD (+/-)				0.06		52.07	
CV (%)				2.9%		7.8%	
N				3		3	

Table 7 Short term stability in cell culture media MCM+(addition of 0.1% SDS to working solutions)

2439-exNA				QC-Low 3.0 ng/mL		QC-High 900.0 ng/mL	
Run ID	**Date**	**Sample No.**	**Time Point**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion +/-15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion +/-15 % of Nominal**
AN400.0-24_04_AB01_Qualibatch WS in LoBind addition of 0.1% SDS to WS	27-Jun-2024	30, 33	0 hours	2.8	93.3%	916.0	101.8%
	27-Jun-2024	31, 34		2.9	96.7%	896.1	99.6%
	27-Jun-2024	32, 35		3.0	100.0%	924.9	102.8%
Mean				2.90	96.7%	912.33	101.4%
SD (+/-)				0.10		14.75	
CV (%)				3.4%		1.6%	
N				3		3	
AN400.0-24_04_AB01_Qualibatch WS in LoBind addition of 0.1% SDS to WS	27-Jun-2024	72, 75	4 hours on ice	2.9	96.7%	920.8	102.3%
	27-Jun-2024	73, 76		2.8	93.3%	925.4	102.8%
	27-Jun-2024	74, 77		3.0	100.0%	903.7	100.4%
Mean				2.90	96.7%	916.63	101.8%
SD (+/-)				0.10		11.43	
CV (%)				3.4%		1.2%	
N				3		3	

Table 7 Short term stability in cell culture media MCM+(addition of 0.1% SDS to working solutions)_continued

2439-exNA				QC-Low 3.0 ng/mL		QC-High 900.0 ng/mL	
Run ID	**Date**	**Sample No.**	**Time Point**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion +/-15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion +/-15 % of Nominal**
AN400.0-24_04_AB01_Qualibatch WS in LoBind addition of 0.1% SDS to WS	27-Jun-2024	51, 54	4 hours at RT	2.6	86.7%	912.0	101.3%
	27-Jun-2024	52, 55		2.6	86.7%	912.9	101.4%
	27-Jun-2024	53, 56		2.5	83.3%	941.2	104.6%
Mean				2.57	85.6%	922.03	102.4%
SD (+/-)				0.06		16.60	
CV (%)				2.2%		1.8%	
N				3		3	

Table 8 Short term stability in cell culture media MCM+: Addition of 0.1% SDS to workings solution and to stability samples

2439-exNA				QC-Low 3.0 ng/mL		QC-High 900.0 ng/mL	
Run ID	**Date**	**Sample No.**	**Time Point**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion +/-15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion +/-15 % of Nominal**
AN400.0-24_04_AB03_Qualibatch WS in LoBind WS with 0.1 % SDS	15.Jul.2024	17, 20	0 hours	3.0	100.0%	1,009.5	112.2%
	15.Jul.2024	18, 21		3.0	100.0%	1,022.9	113.7%
	15.Jul.2024	19, 22		3.1	103.3%	983.8	109.3%
Mean				3.03	101.1%	1,005.40	111.7%
SD (+/-)				0.06		19.87	
CV (%)				1.9%		2.0%	
N				3		3	
AN400.0-24_04_AB03_Qualibatch WS in LoBind WS with 0.1% SDS	15.Jul.2024	31, 34	4 hours on ice	2.7	90.0%	976.0	108.4%
	15.Jul.2024	32, 35		2.6	86.7%	976.0	108.4%
	15.Jul.2024	33, 36		2.8	93.3%	996.3	110.7%
Mean				2.70	90.0%	982.77	109.2%
SD (+/-)				0.10		11.72	
CV (%)				3.7%		1.2%	
N				3		3	

Table 8 Short term stability in cell culture media MCM+: Addition of 0.1% SDS to workings solution and to stability samples_continued

2439-exNA				QC-Low 3.0 ng/mL		QC-High 900.0 ng/mL	
Run ID	**Date**	**Sample No.**	**Time Point**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion +/-15 % of Nominal**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion +/-15 % of Nominal**
AN400.0-24_04_AB03_Qualibatch WS in LoBind with blank matix, addition of 1vol% 10%SDS to final STS-sample (corr.factor=1.01)	15.Jul.2024	24, 27	0 hours	2.4	80.0%	919.0	102.1%
	15.Jul.2024	25, 28		2.4	80.0%	931.0	103.4%
	15.Jul.2024	26, 29		2.4	80.0%	1,004.0	111.6%
Mean				2.40	80.0%	951.33	105.7%
SD (+/-)				0.00		46.00	
CV (%)				0.0%		4.8%	
N				3		3	
AN400.0-24_04_AB03_Qualibatch WS in LoBind with blank matrix, addition of 1vol% 10%SDS to final STS-sample (corr.factor=1.01)	15.Jul.2024	38, 41	4 hours on ice	2.5	83.3%	902.4	100.3%
	15.Jul.2024	39, 42		2.5	83.3%	862.6	95.8%
	15.Jul.2024	40, 43		2.4	80.0%	877.6	97.5%
Mean				2.47	82.2%	880.87	97.9%
SD (+/-)				0.06		20.10	
CV (%)				2.3%		2.3%	
N				3		3	

Underlined values did not meet the acceptance criteria but used to calculate Mean, SD, CV and N

Table 9 Dilution Integrity in cell culture media MCM+

2439-exNA		Dilution 1:1,000 500,000.0 ng/mL	
Run ID	**Sample No.**	**Calculated 2439-exNA Conc. ng/mL**	**Acc. Criterion +/-15 % of Nominal**
Dilution reagent: LC-MS Water			
AN400.0-24_04_AB02	28	257,004.0	51.4%
	29	250,827.9	50.2%
	30	282,117.9	56.4%
	46	383,647.9	76.7%
	47	388,099.0	77.6%
	48	372,940.5	74.6%
Mean		322,439.53	64.5%
SD (+/-)		65,793.38	
CV (%)		20.4%	
N		6	
Dilution reagent: 4 M Urea			
AN400.0-24_04_AB02	49	443,540.0	88.7%
	50	484,491.8	96.9%
	51	537,260.4	107.5%
Mean		488,430.73	97.7%
SD (+/-)		46,984.20	
CV (%)		9.6%	
N		3	
Dilution reagent: cell medium buffer			
AN400.0-24_04_AB02	52	387,166.1	77.4%
	53	472,117.6	94.4%
	54	456,462.6	91.3%
Mean		438,582.10	87.7%
SD (+/-)		45,210.33	
CV (%)		10.3%	
N		3	
Dilution reagent: cell medium buffer/ 1% Tween / 1 mM TCEP			
AN400.0-24_04_AB02	55	438,232.5	87.6%
	56	482,043.0	96.4%
	57	457,331.0	91.5%
Mean		459,202.17	91.8%
SD (+/-)		21,965.11	
CV (%)		4.8%	
N		3	

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6 Records

The following records in addition to the above-mentioned Forms, Master or Spreadsheet Applications are required in this process:

- N/A

7 Change History

Edition, Section(s)	Change	Rationale
Ed 01	New document	Assay description after the method development
Ed 02	Cell medium name MCM+ added and further explanation of hybridization step for study samples inserted	- specification of cell medium MCM+ added in section 3.1 and in data tables in section 4 - explanation of hybridization step for study samples inserted in section 5.4.3